Northeastern University, Khoury College of Computer Science

CS 6120 Natural Language Processing — Lab T14

April 13, 2026 (Week 12)

Agents with an Execution Environment

Objective

The goal of this lab is to set up a self-hosted agent using **OpenClaw**. You will provision a virtual environment, install the OpenClaw execution framework, and bridge it to a Telegram account to allow for remote agentic interactions.

Step 1: Telegram Account Setup

To interact with your agent via messaging, you need an active Telegram account.

- Install the Telegram app on your mobile device or desktop.
- Register using a valid phone number.

Step 2: Provisioning the Execution Environment

OpenClaw requires a Linux environment to run commands and manage tools.

1. Navigate to the **Google Cloud Console (GCP)**.
2. Create a new Virtual Machine (VM) instance (e2-medium is recommended).
3. Ensure you have SSH access to the instance and that outgoing traffic is allowed.

Step 3: Installing OpenClaw

Run the following command in your VM terminal to install the OpenClaw CLI:

```
curl -fsSL https://openclaw.ai/install.sh | bash
```

Step 4: API Configuration & Custom JSON Setup

OpenClaw defaults to OpenAI. However because we have Google credits and for better stability and feature support, we will configure it to use Gemini via a custom JSON configuration.

1. **Obtain API Keys:** There are several open LLM options that you can use, but you'll need an API key to call them. Alternatively, you can use your own if you're far enough in your project. Here are the steps to configure to access an LLM.
   a) Log in to Google Vertex AI, and select API Keys to generate your own.
   b) Log in to the **Google AI Studio** console with your project and create a new Secret Key: Google's AI Studio
   c) You may also need additional keys. There are a handful of sites that provide free API keys for limited usage. One that you can try is: OpenRouter.AI

   You may wish to specify several LLM models, where you can add fallbacks in case you become rate limited.

2. **Review Reference Configuration:** The main file that you create when installing OpenClaw is the json configuration file. You'll need to edit it for any new tool that you'd like to provide it.
   - The file is located at: `.openclaw/openclaw.json`
   - Feel free to take a look at my openclaw.json file. (Don't copy it; it's only a reference.)

3. **Edit the Credentials:** Open the file to insert your specific API keys:
   `nano .openclaw/openclaw.json`

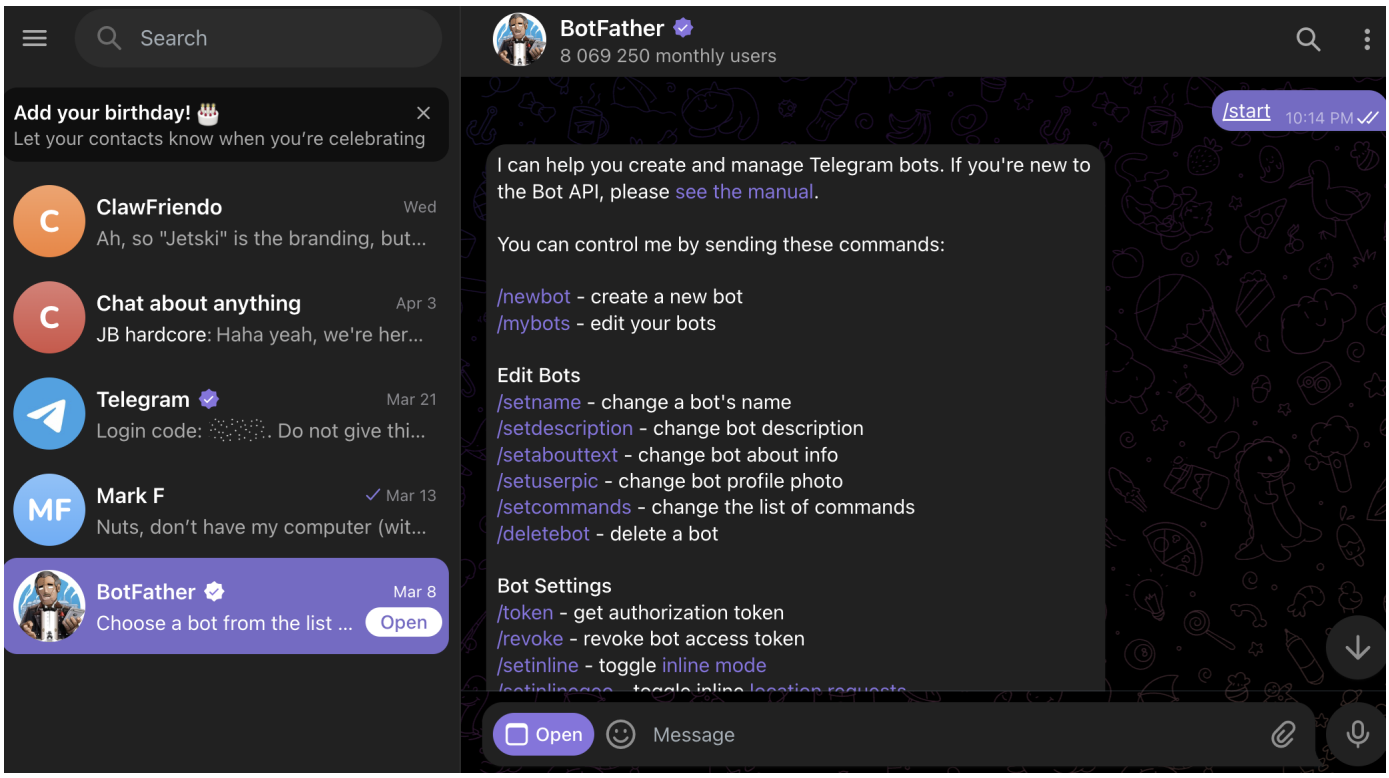
4. **Update the Model Provider:** Inside the file, ensure the `"provider"` is set to `"google"` and paste your key into the `"api_key"` field.

5. **Verify Schema:** OpenClaw is strict about JSON formatting. You can verify your file is valid by running: `openclaw config doctor`

Step 5: Connecting Messaging Channels via @BotFather

To link OpenClaw to Telegram, you must first create a bot identity:

1. Open Telegram and search for **@BotFather**.
2. Send the command `/newbot` and follow the prompts to name your bot.
3. Copy the **API Token** provided by @BotFather.
4. In your VM terminal, run: `openclaw channels add`
5. Select **Telegram** and paste your API Token when prompted.
6. Verify the connection with: `openclaw channels list`



Take a look at your `.openclaw/openclaw.json` file. If you were added to a group, your configuration should look like the following, where the groups are denoted by the numbers starting with 512 below.

```
"channels": {
  "telegram": {
    "enabled": true,
    "dmPolicy": "pairing",
    "botToken": "<your-token>",
    "groups": {
      "-5121088395": {
        "requireMention": false,
        "groupPolicy": "open"
      }
    },
    "groupAllowFrom": [
      "-5121088395"
    ],
    "groupPolicy": "allowlist",
    "streaming": "partial"
  }
},
```

Try chatting with your bot. You may get something that looks like this:

```
OpenClaw: access not configured.
```

```
Your Telegram user id: XXXXXXXX
Pairing code: XXXXXXXX
Ask the bot owner to approve with:
openclaw pairing approve telegram XXXXXXX
```

This just means you will need to pair your OpenClaw account with the telegram account. You can do so with:

```
openclaw pairing approve telegram XXXXXXX
```

Exercise

Once connected, message your new bot on Telegram and ask it: *"Check the disk usage on my server and tell me if I'm running out of space."* Then, join ask course staff to join the group, noting the name of your ClawBot.